DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING						
Course Code: 24CS350	***Course Title: Discrete Mathematical Structures***					
Total Credits: 03 **L-45: T-0: SL and TW-45**	***SDGs addressed: 4, 9, 11, 16,17***		***Course Category: PCC***	***Course Type: Theory***		
Teaching Learning Process		***Total Contact Hours/semester***	***Assessment Weightage and Marks***			
	Lecture	**45**		**CIE**	**SEE**	**Total**
	Tutorial	**-**	***Weightage, %***	***40***	***60***	***100***
	Practical	**-**	***Maximum Marks***	***40***	***60***	***100***
	Self-learning and Term Work	**45**	***Minimum Marks***	***20***	***25***	***45****
	Total	**90**				

Note: *For passing the student has to score a minimum of 45 Marks (CIE+SEE: 20 + 25 or 21 + 24)

Course Prerequisite: Basic mathematics.

Course Objectives:

Sl. No.	Course Objectives
1	Introduce concepts of mathematical logic for analyzing propositions and proving theorems.
2	Work with relations, digraphs and investigate their properties.
3	Introduce the concepts of coding, decoding theory, trees and graphs.

Course Outcomes (COs)

CO	Course Outcomes	Highest Level of Cognitive Domain	Working Knowledge
CO1	Use the knowledge needed to test the logic of the program.	L3	WK2, WK3
CO2	Demonstrate the concept of relations and be able to determine their properties.	L3	WK2, WK3
CO3	Explain and apply decode group codes based on the various methods.	L3	WK2, WK4
CO4	Identify different types of graphs such as Isomorphism, Bipartite, planar, Hamiltonian paths and cycles, Euler trails and circuits based on their properties.	L3	WK2, WK4
CO5	Explain different types of trees and apply tree traversal techniques like preorder and postorder.	L3	WK2, WK3

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

Course Content / Syllabus:

UNIT No.	Content	Lecture Hours
1	**Fundamental of Logic:** Basic Connectives and Truth Tables, Logical Equivalence: The Laws of Logic, Logical Implication: Rules of Inference, The Use of Quantifiers, Quantifiers.	09

	Self-Learning and Team Work*: • Practice creating and verifying truth tables independently. • Explore real-life applications of logical equivalence (digital circuits, program verification). • Peer evaluation of logical arguments to strengthen reasoning.	
2	**Relations:** Cartesian Products and Relations, Properties of relations, Computer Recognition: Zeros- One Matrices and Directed Graphs, Partial; orders: Hasse diagrams, Equivalence Relations and Partitions, Lattices. **Self-Learning and Team Work*:** • Study and classify relations using examples from databases or social networks. • Explore lattice applications in decision-making systems individually.	09
3	**Coding Theory:** Elements of coding theory and Hamming Metric, generation of codes using Parity check and Generator matrices. **Self-Learning and Team Work*:** • Learn to decode and encode messages using Hamming codes or generator matrices through individual assignments. • Implement group coding exercises to simulate error detection and correction in a communication channel.	09
4	**Graph theory:** Definitions and Examples, Subgraphs, Complements, and Graph Isomorphism, Vertex degree, Euler trail and circuits, Planar Graphs, Hamiltonian Paths and cycles. **Self-Learning and Team Work*:** • Independently identify graph types and verify properties using small examples. • Use software tools like NetworkX (Python) to explore graph isomorphism. • Collaborative exercises on Euler/Hamiltonian paths and planar graph identification.	09
5	**Trees:** Definitions, Properties and Examples, Rooted trees, trees and sorting weighted trees and prefix Codes. **Self-Learning and Team Work*:** • Practice tree traversals (preorder, postorder) with programming tasks. • Learn about applications in data compression (prefix codes) individually. • Team-based exploration of weighted trees for optimization problems (like Huffman coding).	09

NOTE: *Self-Learning and Term Work are evaluated only through continuous internal evaluation (CIE) and will not be assessed during the Semester End Examination (SEE).

Text Book:

Sl. No.	Author/s	Title	Publisher Details
1	Ralph. P. Grimaldi, B.V. Ramana	Discrete and Combinatorial Mathematics	5th Edition, Pearson Education, 2013.

Reference Books:

Sl. No.	Author/s	Title	Publisher Details
1	Kenneth. H. Rosen	Discrete Mathematical Structures Theory and Application	5th Edition, PHI/Pearson, Education, 2004.
2	Kolman, Busby and Ross	Discrete Mathematical Structures	4th Edition, Prentice Hall of India Pvt. Ltd, 2009.
3	Dr. D.S. Chandrashekar	Discrete Mathematical Structures	4thEdition Prism Books Private Limited, 2013.
4	Dr. S.K. Yadav	Discrete Mathematical Structures	Ane Books Pvt. Ltd.

Web Resources:

Sl. No.	Web Links
1	https://nptel.ac.in/courses/106/106/106106094/
2	https://nptel.ac.in/courses/106106183/

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	3	-	-	-	-	-	-	-	-	3	-	3
CO2	3	3	3	-	-	-	-	-	-	-	-	3	-	3
CO3	3	3	3	-	-	-	-	-	-	-	-	3	-	3
CO4	3	3	3	-	-	-	-	-	-	-	-	3	-	3
CO5	3	3	3	-	-	-	-	-	-	-	-	3	-	3

High – 3, Medium – 2, Low – 1